Name: _____.

Instructions

- You may bring one standard 8.5x11 sheet of paper with handwritten notes.
- Unless explicitly requested by a problem, do not include sketches as part of your proof.

Question:	1	2	3	4	Total
Points:	25	25	25	25	100
Score:					

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1. (25 points) Let $P = (u = v_0, v_1, \ldots, v_k = v)$, $k \geqslant 1$, be a u–v geodesic in a connected graph G. Prove that $d(u, v_i) = i$ for each integer i with $1 \leqslant i \leqslant k$.

Solution: Note for later use that $d(u, v) = k$, because $P = (u = v_0, v_1, \ldots, v_k = v)$ is a geodesic.

Fix any $i \in \{0, \ldots, k\}$. Since$(u = v_0, v_1, \ldots, v_i)$ is a u–v_i path of length i, we know that $d(u, v_i) \leqslant i$. Suppose for the sake of contradiction that $d(u, v_i) = \ell < i$; hence there is a u–v_i geodesic $(u = w_0, \ldots, w_\ell = v_i)$. But then $(u = w_0, \ldots, w_\ell = v_i, v_{i+1}, \ldots, v_k = v)$ is a u–v walk of length $\ell + (k - i)$. Theorem 1.6 then implies that there is a u–v path of length at most $\ell + (k - i) < k$, contradicting the previously established fact that $d(u, v) = k$.

2. (25 points) Let G be a connected graph and consider a function $f\colon V(G) \to X$ where X is some arbitrary set. Prove that if f is *not* a constant function, then there is an edge $uv \in E(G)$ such that $f(u) \neq f(v)$.

   Deduce that if G is not regular, then G contains adjacent vertices u and v such that $\deg u \neq \deg v$.

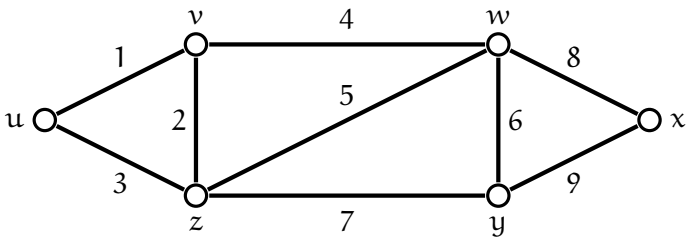
Solution: Let us show the contrapositive statement. Thus we assume that for any pair of vertices u, v in G, if $uv \in E(G)$ then $f(u) = f(v)$. To see that f must be constant, fix any vertex u^* in G. Now, for any other vertex v, since G is connected, there is a u^*–v path in G, say $(u^* = v_0, v_1, \ldots, v_n = v)$. This means that $v_i v_{i+1} \in E(G)$ for each $i \in \{0, 1, \ldots, n-1\}$, thus $f(v_i) = f(v_{i+1})$ for any such i. Putting these n equations together, we get

$$f(u^*) = f(v_0) = f(v_1) = f(v_2) = \cdots = f(v_{n-1}) = f(v_n) = f(v).$$

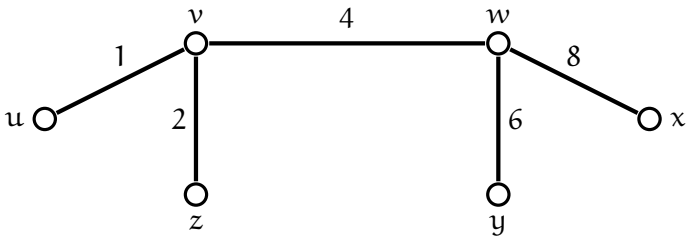
Thus $f(v) = f(u^*)$ for any $v \in V(G)$, and f is constant.

In case G is not regular, the degree function $\deg\colon V(G) \to \mathbb{N}$ is not constant. Hence we deduce from the statement above that there is an edge $uv \in E(G)$ such that $\deg u \neq \deg v$. □

3. (25 points) Apply both Kruskal's and Prim's Algorithms to find a minimum spanning tree in the following graph. In each case, show how this tree is constructed.



Solution:



4. (25 points) For this problem, you need to select **one of the two statements below**.

   (a) Let G be a self-complementary graph of order $n = 4k$, where $k \geqslant 1$. Let $U = \{v : \deg v \leqslant n/2\}$ and $W = \{v : \deg v \geqslant n/2\}$. Prove that if $|U| = |W|$, then G contains no vertex v such that $\deg v = n/2$.
   Hint: $\overline{G}$ contains at least $|W|$ many vertices with $\overline{G}$-degree strictly smaller than $n/2$.

   (b) Prove that a 3-regular graph G has a cut-vertex if and only if G has a bridge.

Solution:

(a) Let $V = V(G) = V(\overline{G})$. Recall that for any $v \in V$, we have $\deg_G v = n - 1 - \deg_{\overline{G}} v$.

Note that if $v \in W$, then $\deg_G v \geqslant n/2$, thus $\deg_{\overline{G}} v = n-1-\deg_G v \leqslant n-1-n/2 = n/2-1$. Hence $\overline{G}$ contains at least $|W|$ many vertices with $\overline{G}$-degree strictly smaller than $n/2$.

Since $G \cong \overline{G}$, we deduce that also G contains $|W|$ many vertices with G-degree strictly smaller than $n/2$. Since $|W| = |U| = |\{v : \deg v \leqslant n/2\}|$, this means that any vertex v in U must satisfy $\deg_G v < n/2$. Hence G contains no vertices of degree $n/2$.

(b) Since a graph is 3-regular if and only if all its components are 3-regular, by reducing the argument to a component, we can assume that G is connected.

Assume first G has a bridge $e = uv$. Then, by Theorem 5.1, u and v are cut vertices, because both have degree $\geqslant 2$.

Conversely, let v be a cut-vertex. Since v has exactly 3 neighbors, $G - v$ has either 2 or 3 components. If $G - v$ has 2 components one of these has two neighbors and the other has one. If $G - v$ has 3 components, then each of these contains only one neighbor. In either case, we see that there is a component H of $G - v$ with exactly one neighbor of v, which we denote by u.

We show that $e = uv$ is a bridge of G. Indeed, we show that e is the unique u–v path in G. For a contradiction, if $(u = v_0, v_1, \ldots, v_n = v)$ is a u–v path in G of length $n \geqslant 2$, then the vertices $v_1, \ldots, v_{n-1}$ must be in $V(H)$, since they connect to $u = v_0$ without passing trough $v = v_n$. But then v_{n-1} is a vertex in H such that $v_{n-1}v$ is an edge in G, which contradicts that u is the unique G-neighbor of v in H. Hence e is a bridge.

Note: it is not true that two incident cut-vertices form a bridge.